A MINI-PROJECT REPORT
ON

Car Price Prediction: A Machine Learning-Based Web Application

Submitted by

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UNDER THE GUIDANCE OF
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towards the partial fulfillment of

Third Year Under Graduate Course

in

Computer Engineering of

SAVITRIBAI PHULE PUNE UNIVERSITY in the academic year 2026-2027.



DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that Mini-project Report
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Project Guide **HOD**

Acknowledgement

I would like to express my sincere gratitude to my project guide for providing valuable guidance and support throughout the development of the Car Price Prediction project.

I am thankful to the faculty members of the Computer Engineering department at Ajeenkya D. Y. Patil School Of Engineering Lohegaon Pune for providing the academic knowledge and resources required to complete this project.

I would also like to acknowledge the developers of Python and its supporting libraries, which provide essential tools for data processing, model development, evaluation, and application creation.

I am grateful to the open-source community and the platforms that support sharing datasets, notebooks, and source code. These resources helped me understand the practical implementation of machine-learning techniques.

Finally, I would like to thank my family and friends for their encouragement and support throughout the project.

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Abstract

Car Price Prediction is a machine-learning web application developed to estimate the resale price of a used car based on its characteristics. The price of a used car depends on several factors, including the distance travelled, fuel type, selling type, transmission, previous ownership, and age of the vehicle.

Estimating the correct price manually can be difficult because different vehicles have different specifications and market values. This project uses machine-learning techniques to analyse historical car data and predict an estimated price based on the information entered by the user.

The application provides an interactive web interface where users can enter vehicle details through input fields and dropdown menus. The interface includes fields such as driven kilometres, fuel type, selling type, transmission, owner category, and car age. It also provides links to the GitHub repository and Kaggle notebook for accessing the project's source code and related work.

The project involves dataset analysis, data preprocessing, feature selection, model training, and model evaluation. After training, the model can use the supplied vehicle information to generate a price estimate.

The main objective is to demonstrate the practical application of machine learning in the automobile resale market. The application can help users obtain a preliminary price estimate, although the final market value may differ depending on vehicle condition, location, demand, and other factors not represented in the dataset.

Keywords: Machine Learning, Car Price Prediction, Regression, Python, Data Analysis, Web Application, Used-Car Valuation.

Introduction

The used-car market provides an affordable alternative to purchasing new vehicles. Many people buy and sell pre-owned cars, making accurate price estimation an important part of the buying and selling process.

The price of a car depends on several characteristics. A newer vehicle may generally have a higher resale value than an older vehicle of a similar type. The distance travelled, fuel type, transmission, ownership history, and selling category can also influence the price.

Traditional methods of car valuation often depend on manual calculations, depreciation estimates, and comparisons with similar vehicles. These methods may not consistently consider all available information.

Machine learning provides a data-driven way to address this problem. By analysing historical car records, a model can learn relationships between vehicle characteristics and their corresponding prices.

This project implements a machine-learning-based web application that accepts car details and generates an estimated resale price. The user interacts with the application through a simple form instead of having to understand the underlying machine-learning algorithm.

The project combines data analysis, model training, and web application development to demonstrate how machine learning can solve a practical estimation problem.

Problem Statement

Estimating the resale price of a used car is challenging because the price depends on multiple factors, and their effects may differ between vehicles.

Manual estimation may produce inconsistent results and can require considerable time to compare similar cars.

The problem addressed by this project is to develop a web application that uses machine learning to predict the estimated resale price of a car based on its available characteristics.

The system should accept vehicle details, process the information using the trained model, and present a predicted price through a simple user interface.

Objectives

The main objectives of the project are:

1. To develop a machine-learning model for predicting used-car prices.
2. To analyse historical car-price data and understand the important vehicle attributes.
3. To preprocess the dataset and prepare it for model training.
4. To train and evaluate a suitable regression algorithm.
5. To develop a user-friendly web application for entering car details.
6. To generate a price estimate from the information provided by the user.
7. To make the project source code and notebook accessible through repository links.
8. To understand the practical use of machine learning in automotive price estimation.
9.

Scope of the Project

The project focuses on estimating the resale price of a used car using the vehicle attributes supported by the application.

As shown in the application screenshot, the user can provide details such as:

- Driven kilometres
- Fuel type
- Selling type
- Transmission
- Owner category
- Car age

The application processes these inputs through the trained model and produces an estimated price.

The project is primarily intended for educational purposes and preliminary price estimation. It does not perform a physical inspection of the vehicle or independently verify its maintenance history, accident records, or legal documents.

The actual prediction quality depends on the dataset, the model used, and the similarity between the input vehicle and the vehicles represented in the training data.

Existing System

In the traditional system, the resale price of a car is generally estimated through manual inspection, dealer experience, depreciation calculations, and comparisons with similar vehicles.

A buyer or seller may check online listings and adjust the expected price according to the car's age, mileage, brand, and condition.

However, this approach has certain limitations:

- The estimated price may vary between individuals.
- Manual comparisons can be time-consuming.
- Different vehicle characteristics may not be evaluated consistently.
- Static depreciation calculations may not reflect current market demand.
- It can be difficult to analyse a large amount of historical vehicle data manually.

Proposed System

The proposed system uses a trained machine-learning model to estimate a used car's price from the characteristics entered by the user.

The application provides a web interface with input fields and dropdown menus. The user enters the vehicle details, and the system passes the information to the prediction model.

The model processes the input features and generates an estimated price. The result is then presented to the user through the application.

The system also includes links to the GitHub repository and Kaggle notebook, allowing users to access the related project resources.

Working process

1. Prepare and load the car-price dataset.
2. Inspect and preprocess the data.
3. Analyse the relationships between vehicle features and prices.
4. Train the machine-learning model.
5. Evaluate the model's predictive performance.

6. Connect the trained model to the web application.
7. Accept vehicle information through the user interface.
8. Display the estimated car price.

Technologies Used

The following technologies are suitable for this project. Retain only those that were actually used in your implementation.

Technology	Purpose
Python	Main programming language
Pandas	Loading and processing the dataset
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Exploratory data analysis
Scikit-learn	Model training and evaluation
Streamlit framework	Building the interactive interface, if used
GitHub	Hosting the project source code
Kaggle	Hosting or presenting the notebook and dataset-related work

Python supports the development of the complete machine-learning workflow, while the associated libraries simplify data analysis and model development.

Theoretical Background

10.1 Machine Learning

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and use them to make predictions.

In this project, the model learns from historical vehicle records containing car features and known prices. It uses the learned relationships to estimate the price of another vehicle.

10.2 Supervised Learning

Supervised learning uses labelled data. Each training record contains input information and a known output.

In the Car Price Prediction project, vehicle characteristics are the inputs, and the recorded car price is the output that the model learns to predict.

10.3 Regression

Regression is a machine-learning technique used to predict continuous numerical values.

Since car price is a numerical value, this project is a regression problem. The model learns the relationship between the vehicle attributes and the price.

10.4 Data Preprocessing

Data preprocessing prepares raw data for machine learning. It may include handling missing values, removing inappropriate duplicate records, correcting data types, and converting categorical information into a suitable numerical representation.

Proper preprocessing helps the model work with consistent and meaningful inputs.

10.5 Model Training

During model training, the algorithm analyses the training dataset and learns patterns connecting vehicle features with car prices.

The trained model is then evaluated using data that was not used to fit it. This helps determine whether the model can make useful predictions for unseen vehicles.

10.6 Model Evaluation

Model evaluation measures how closely the predicted prices match the actual prices.

Common regression metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score.

Lower MAE and RMSE generally indicate smaller prediction errors, while R^2 helps assess how well the model explains variation in the target values.

The actual metric values should be obtained from the project's evaluation output.

Machine-Learning Model

The project uses a regression model to estimate the car's price. The exact algorithm should be named according to the implementation in the GitHub repository or Kaggle notebook.

Common algorithms suitable for this task include Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression.

Linear Regression provides a simple baseline. Decision Tree Regression can learn nonlinear patterns, while Random Forest and Gradient Boosting combine multiple trees to model more complex relationships.

The best algorithm should be selected based on its measured performance on unseen data. The report should not claim that a particular algorithm was used unless it is confirmed by the project code.

Web Application Design

The web application is the user-facing component of the project. Its purpose is to make the trained machine-learning model accessible through an interactive form.

According to the supplied screenshot, the interface contains the following elements.

14.1 Application Title

The application displays the title Car Price Prediction, with a subtitle describing it as a machine-learning web application for predicting used-car prices.

This helps users understand the purpose of the application immediately.

14.2 GitHub Repository Link

The GitHub Repository link provides access to the project's source code, subject to the repository's access permissions.

It allows developers and evaluators to inspect the implementation, dependencies, and project structure.

14.3 Kaggle Notebook Link

The Kaggle Notebook link provides access to the associated notebook, if available. The notebook can document data analysis, preprocessing, model development, and evaluation.

14.4 Driven Kilometres

This field accepts the total distance travelled by the vehicle. The screenshot shows an example value of 50,000 kilometres.

The model can use mileage-related information to learn how vehicle usage relates to historical prices.

14.5 Fuel Type

This dropdown allows the user to select the fuel category. The screenshot shows Petrol as the selected example.

Other options depend on the categories supported by the trained model.

14.6 Selling Type

This field represents the selling category. The screenshot shows Dealer as the selected example.

The available categories and their meanings must follow the dataset used in training.

14.7 Transmission

This dropdown allows the user to specify the transmission type. The screenshot shows Manual as the selected example.

The application should use the same category representation as the model's preprocessing pipeline.

14.8 Owner

This dropdown represents the previous-owner category. The screenshot shows the value 0.

The interpretation of each value should follow the original dataset rather than assuming that the number directly represents a natural-language owner count.

14.9 Car Age

This field allows the user to specify the age of the car. Car age can help the model learn depreciation-related patterns from historical data.

The precise age calculation should match the project's implementation.

14.10 Price Prediction Output

After the user enters the required details and submits the form, the application processes the inputs through the trained model and displays the estimated price.

The supplied screenshot does not show the prediction result, so a screenshot of the actual output should be added to the final report.

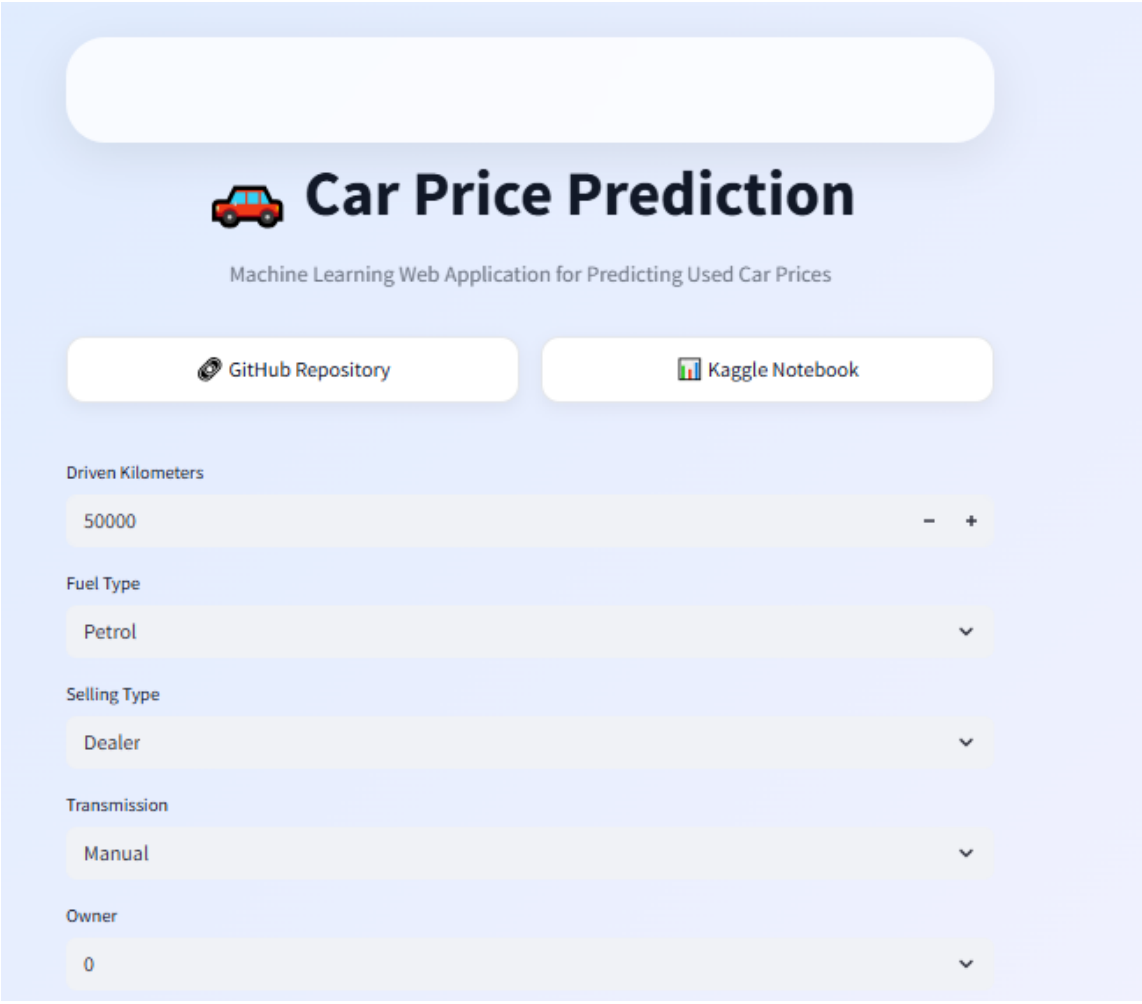
Results and Discussion

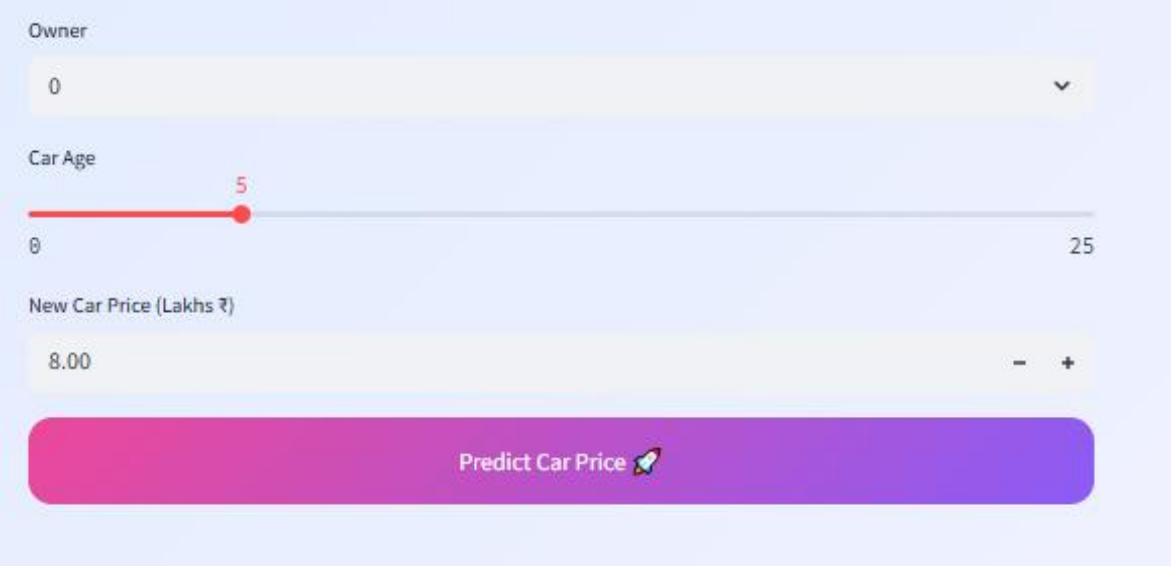
The project aims to provide a convenient way to estimate the resale price of a car through a web application.

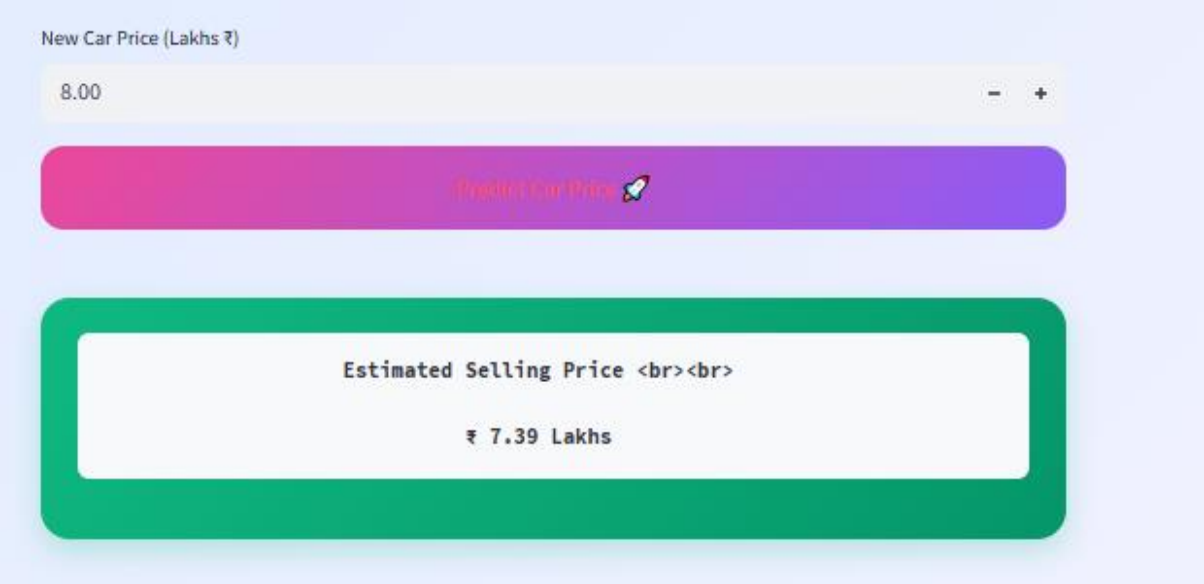
The user interface simplifies the process by allowing vehicle characteristics to be entered through input fields and dropdown menus. The trained regression model then processes those characteristics and generates a numerical estimate.

The quality of the predictions should be evaluated using the actual test-set results. The report should include the selected algorithm, Mean Absolute Error, Root Mean Squared Error, and R^2 score, where available.

The model's performance depends on the quality of the dataset, the features included in training, and how well the training examples represent the vehicle being evaluated.







Live Link : https://carprediction-996x.onrender.com/

Future Scope

The project can be improved in several ways.

Improved dataset: More recent and representative vehicle data could help the model reflect changing market conditions.

Additional features: Vehicle condition, service history, location, and accident records could be incorporated where reliable data is available.

Model comparison: Different regression algorithms could be compared to determine which provides the best performance for the dataset.

Improved interface: The application could provide clearer input validation, prediction explanations, and a more detailed result summary.

Deployment: The web application could be deployed to an appropriate hosting platform to make it accessible to users.

Model maintenance: The system could be updated periodically using new data, with performance checks before deploying an updated model.

Conclusion

The Car Price Prediction project demonstrates how machine learning can be applied to a practical problem in the used-car market.

The project combines data preprocessing, exploratory analysis, regression modelling, model evaluation, and a web-based interface. Users can enter vehicle characteristics such as driven kilometres, fuel type, selling type, transmission, ownership category, and car age to obtain an estimated price.

The application makes the prediction process more accessible by providing an interactive interface and links to the associated project resources.

The project also demonstrates important concepts in machine learning, including supervised learning, regression, feature preparation, training, testing, and evaluation.

Although the system can support preliminary price estimation, its reliability depends on the quality of the data and the model's measured performance. Future improvements can include better datasets, additional vehicle attributes, model comparison, and application deployment.

Overall, the project provides practical experience in developing a machine-learning web application and illustrates how data-driven methods can support used-car valuation.

References

1. Scikit-learn Developers, Scikit-learn Documentation. https://scikit-learn.org/
2. Pandas Development Team, Pandas Documentation. https://pandas.pydata.org/docs/
3. NumPy Developers, NumPy Documentation. https://numpy.org/doc/
4. Matplotlib Development Team, Matplotlib Documentation. https://matplotlib.org/stable/
5. Seaborn Developers, Seaborn Documentation. https://seaborn.pydata.org/
6. James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J., An Introduction to Statistical Learning: With Applications in Python. Springer, 2023. https://www.statlearning.com/